SLAB DESIGN CALCULATIONS

Project Name	:	G+2 RB SLAB DESIGN @F1
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Slab-1-5.029 m.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45

NOTES:

1. Slab is designed as per 1 M width of panel.
2. Density of concrete assumed as 25 kN/Cum.
3. Maximum spacing of Main steel is smaller of (3 x Deff and 300 mm)
4. Maximum spacing of Distribution steel is smaller of (5 x Deff and 450 mm)
5. All 6 mm diameter bars assumed of Fe250 grade with Fy=250 N/sq.mm

Definitions Of Terms:

1.	α	=	Coefficient of Thermal Expansion
2.	acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
3.	εm	=	Average steel strain at level considered
4.	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
5.	ρact	=	Percentage steel provided
6.	ρcrit	=	Percentage steel critical as per code
7.	BM	=	Bending moment in kNm
8.	D	=	Overall depth of slab in mm
9.	Deff	=	Effective depth of slab in mm
10.	Ec	=	Modulus of elasticity of concrete
11.	Es	=	Modulus of elasticity of steel
12.	Fck	=	Characteristic compressive strength of concrete cube in N/sqmm
13.	Fst	=	Stress in steel
14.	Fy	=	Yield Stress Of Steel in N/sqmm
15.	Mu	=	Factored Bending moment in kNm
16.	Sp	=	Spacing Between bars at outer most layer
17.	Tc	=	Permissible Shear stress in concrete in N/sqmm
18.	Temp1	=	Peak Hydration temperature
19.	Temp2	=	Seasonal Temperature Variations
20.	Tv	=	Shear stress due to Vu in N/sqmm
21.	V	=	Shear force in kN
22.	Vu	=	Factored shear force in kN
23.	Wcr	=	Surface Crack Width

Code References :
IS 456

	ELEMENT		CLAUSE / TABLE
1.	Moment Coeffs For 2-Way Slabs	:	Greschoff-Rankine Formulation and yield line theory
2.	Ptmin	:	26.5.2.1
3.	Ptcalculated	:	38
4.	Shear Strength Of Concrete	:	40.3
5.	Modification Factor for Shear	:	40.2.1.1
6.	Max Nominal Shear Stress	:	40.2.3
7.	Basic Span To Depth Ratio	:	23.2.1
8.	Tensile Steel Factor	:	23.2.1
9.	Compression Steel Factor	:	23.2.1
10.	Flange correction factor	:	23.2.1
11.	spacing between main bars	:	26.3.3
12.	spacing between distribution bars	:	26.3.3
13.	crack width calculation	:	Annex-F
14.	cracking	:	Clause 35.3.2

BS 8007 (For Early / Initial Thermal Cracking)

	ELEMENT		CLAUSE / TABLE
1.	Surface Zone for suspended slab	:	Figure A.1
2.	Surface Zone for ground slab	:	Figure A.2
3.	Factors for the calculation of minimum reinforcement	:	Table A.1

Slab No. :	**S1**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.990	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	125.000	mm

Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	Two Adjacent Edges Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.050	
BM	=	5.123	kNm
Mu = 1.5 X BM	=	7.685	kNm
Area Of Reinforcement Required (Mu)	=	166.746	sqmm/m
Area Of Reinforcement Required (Mu + Defl.)	=	216.746	sqmm/m
Reinforcement Provided	=	T8 @ 230 C/C	
	=	219.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.067	
BM	=	6.831	kNm
Mu = 1.5 X BM	=	10.246	kNm
Area Of Reinforcement Required (Mu)	=	225.360	sqmm/m
Additional Area Of Reinforcement Required	=	116.987	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.034	
BM	=	3.461	kNm
Mu = 1.5 X BM	=	5.192	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (BM + Defl.)	=	195.000	sqmm/m
Reinforcement Provided	=	T8 @ 255 C/C	
	=	197.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.045	
BM	=	4.615	kNm
Mu = 1.5 X BM	=	6.923	kNm
Area Of Reinforcement Required (Mu)	=	167.652	sqmm/m
Additional Area Of Reinforcement Required	=	70.152	sqmm/m

Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.492	
V = SF1 x TL x Lx	=	12.774	kN
Vu = 1.5 X V	=	19.161	kN
Shear Stress, Tv	=	0.192	N/sqmm
Nominal Shear Stress, Tc	=	0.349	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.453	
	>	0.192	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.400	
V = SF2 x TL x Lx	=	10.388	kN
Vu = 1.5 X V	=	15.582	kN
Shear Stress, Tv	=	0.173	N/sqmm
Nominal Shear Stress, Tc	=	0.321	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.417	
	>	0.173	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.678	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.68 x 1.00 x 1.00	
	=	43.629	
Lx/d	=	39.200	
	<	43.629	Slab Is Safe In Deflection

Slab No. :	**S2**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.990	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	One Short Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -			
Moment Coefficient	=	0.038	
BM	=	3.886	kNm
Mu = 1.5 X BM	=	5.828	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (Mu + Defl.)	=	175.000	sqmm/m
Reinforcement Provided	=	T8 @ 285 C/C	
	=	176.000	sqmm/m
Short Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.051	
BM	=	5.181	kNm
Mu = 1.5 X BM	=	7.771	kNm
Area Of Reinforcement Required (Mu)	=	168.698	sqmm/m
Additional Area Of Reinforcement Required	=	81.198	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m
Long Span Positive Moment At Midspan -			
Moment Coefficient	=	0.028	
BM	=	2.800	kNm
Mu = 1.5 X BM	=	4.199	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (BM + Defl.)	=	172.500	sqmm/m
Reinforcement Provided	=	T8 @ 290 C/C	
	=	173.000	sqmm/m
Long Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.037	
BM	=	3.733	kNm
Mu = 1.5 X BM	=	5.599	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of	=	63.750	sqmm/m

Reinforcement Required			
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m
Distribution Reinforcement @ 0.12% -			
Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.435	
V = SF1 x TL x Lx	=	11.286	kN
Vu = 1.5 X V	=	16.930	kN
Shear Stress, Tv	=	0.169	N/sqmm
Nominal Shear Stress, Tc	=	0.307	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.399	
	>	0.169	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	9.349	kN
Vu = 1.5 X V	=	14.024	kN
Shear Stress, Tv	=	0.156	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.156	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.593	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.59 x 1.00 x 1.00	
	=	41.427	
Lx/d	=	39.200	
	<	41.427	Slab Is Safe In Deflection

Slab No. :	**S3**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.990	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Dead Load,DL	=	4.625	kN/sqm
Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	1-Way	
End Conditions	=	End Span	
Load Combination	=	1.5 DL + 1.5 LL	

Design Moments:

Short Span Positive Moment at Mid Span -

Moment Coefficient(DL), M1	=	0.083	
Moment Coefficient(LL), M2	=	0.100	
BM = M1 x DL x Lx x Lx + M2 x LL x Lx x Lx	=	1.099	kNm
Mu = 1.5 X BM	=	1.648	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Reinforcement -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Short Span Negative Moment at Continuous Support -

Moment Coefficient(DL), M3	=	0.100	
Moment Coefficient(LL), M4	=	0.111	
BM = M3 x DL x Lx x Lx + M4 x LL x Lx x Lx	=	1.285	kNm
Mu = 1.5 X BM	=	1.928	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

All Other Sections provide Minimum Steel @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Shear Coefficient(DL), SF1	=	0.400	
Shear Coefficient(LL), SF2	=	0.450	

V = SF1 x DL x Lx + SF2 x LL x Lx	=	5.446	kN
Vu = 1.5 X V	=	8.169	kN
Shear Stress, Tv	=	0.082	N/sqmm
Nominal Shear Stress, Tc	=	0.291	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.378	
	>	0.082	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26.000	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.516	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.52 x 1.00 x 1.00	
	=	39.414	
Lx/d	=	13.700	
	<	39.414	Slab Is Safe In Deflection

Slab No. :	**S4**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	5.070	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	One Long Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.047	
BM	=	4.739	kNm
Mu = 1.5 X BM	=	7.108	kNm

Area Of Reinforcement Required (Mu)	=	153.774	sqmm/m
Area Of Reinforcement Required (Mu + Defl.)	=	178.774	sqmm/m
Reinforcement Provided	=	T8 @ 280 C/C	
	=	180.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.062	
BM	=	6.318	kNm
Mu = 1.5 X BM	=	9.477	kNm
Area Of Reinforcement Required (Mu)	=	207.589	sqmm/m
Additional Area Of Reinforcement Required	=	118.202	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.028	
BM	=	2.800	kNm
Mu = 1.5 X BM	=	4.199	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (BM + Defl.)	=	172.500	sqmm/m
Reinforcement Provided	=	T8 @ 290 C/C	
	=	173.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.037	
BM	=	3.733	kNm
Mu = 1.5 X BM	=	5.599	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	63.750	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.468	
V = SF1 x TL x Lx	=	12.154	kN
Vu = 1.5 X V	=	18.231	kN
Shear Stress, Tv	=	0.182	N/sqmm
Nominal Shear Stress, Tc	=	0.336	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.437	
	>	0.182	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	9.349	kN
Vu = 1.5 X V	=	14.024	kN
Shear Stress, Tv	=	0.156	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.156	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.566	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.57 x 1.00 x 1.00	
	=	40.709	
Lx/d	=	39.200	
	<	40.709	Slab Is Safe In Deflection

Slab No. :	**S5**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	5.070	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	Interior Panel	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.034	
BM	=	3.508	kNm
Mu = 1.5 X BM	=	5.263	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (Mu + Defl.)	=	175.000	sqmm/m
Reinforcement Provided	=	T8 @ 285 C/C	
	=	176.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.046	
BM	=	4.678	kNm
Mu = 1.5 X BM	=	7.017	kNm
Area Of Reinforcement Required (Mu)	=	151.736	sqmm/m
Additional Area Of Reinforcement Required	=	64.236	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.024	
BM	=	2.443	kNm
Mu = 1.5 X BM	=	3.665	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (BM + Defl.)	=	172.500	sqmm/m
Reinforcement Provided	=	T8 @ 290 C/C	
	=	173.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.032	
BM	=	3.258	kNm
Mu = 1.5 X BM	=	4.887	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of	=	63.750	sqmm/m

Reinforcement Required			
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m
Distribution Reinforcement @ 0.12% -			
Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.409	
V = SF1 x TL x Lx	=	10.613	kN
Vu = 1.5 X V	=	15.920	kN
Shear Stress, Tv	=	0.159	N/sqmm
Nominal Shear Stress, Tc	=	0.292	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.380	
	>	0.159	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.330	
V = SF2 x TL x Lx	=	8.570	kN
Vu = 1.5 X V	=	12.855	kN
Shear Stress, Tv	=	0.143	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.143	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.593	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.59 x 1.00 x 1.00	
	=	41.427	
Lx/d	=	39.200	
	<	41.427	Slab Is Safe In Deflection

Slab No. :	**S6**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	5.070	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Dead Load,DL	=	4.625	kN/sqm
Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	1-Way	
End Conditions	=	End Span	
Load Combination	=	1.5 DL + 1.5 LL	

Design Moments:

Short Span Positive Moment at Mid Span -

Moment Coefficient(DL), M1	=	0.083	
Moment Coefficient(LL), M2	=	0.100	
BM = M1 x DL x Lx x Lx + M2 x LL x Lx x Lx	=	1.099	kNm
Mu = 1.5 X BM	=	1.648	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Reinforcement -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Short Span Negative Moment at Continuous Support -

Moment Coefficient(DL), M3	=	0.100	
Moment Coefficient(LL), M4	=	0.111	
BM = M3 x DL x Lx x Lx + M4 x LL x Lx x Lx	=	1.285	kNm
Mu = 1.5 X BM	=	1.928	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

All Other Sections provide Minimum Steel @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Shear Coefficient(DL), SF1	=	0.400	
Shear Coefficient(LL), SF2	=	0.450	

V = SF1 x DL x Lx + SF2 x LL x Lx	=	5.446	kN
Vu = 1.5 X V	=	8.169	kN
Shear Stress, Tv	=	0.082	N/sqmm
Nominal Shear Stress, Tc	=	0.291	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.378	
	>	0.082	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26.000	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.516	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.52 x 1.00 x 1.00	
	=	39.414	
Lx/d	=	13.700	
	<	39.414	Slab Is Safe In Deflection

Slab No. :	**S7**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.500	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	One Long Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.039	
BM	=	3.993	kNm
Mu = 1.5 X BM	=	5.990	kNm

Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (Mu + Defl.)	=	175.000	sqmm/m
Reinforcement Provided	=	T8 @ 285 C/C	
	=	176.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.052	
BM	=	5.324	kNm
Mu = 1.5 X BM	=	7.986	kNm
Area Of Reinforcement Required (Mu)	=	173.564	sqmm/m
Additional Area Of Reinforcement Required	=	86.064	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.028	
BM	=	2.800	kNm
Mu = 1.5 X BM	=	4.199	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (BM + Defl.)	=	172.500	sqmm/m
Reinforcement Provided	=	T8 @ 290 C/C	
	=	173.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.037	
BM	=	3.733	kNm
Mu = 1.5 X BM	=	5.599	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	63.750	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.419	
V = SF1 x TL x Lx	=	10.886	kN
Vu = 1.5 X V	=	16.329	kN
Shear Stress, Tv	=	0.163	N/sqmm
Nominal Shear Stress, Tc	=	0.311	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.404	
	>	0.163	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	9.349	kN
Vu = 1.5 X V	=	14.024	kN
Shear Stress, Tv	=	0.156	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.156	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.593	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.59 x 1.00 x 1.00	
	=	41.427	
Lx/d	=	39.200	
	<	41.427	Slab Is Safe In Deflection

Slab No. :	**S8**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.500	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	Interior Panel	

Design Moments:

Short Span Positive Moment At Midspan -			
Moment Coefficient	=	0.030	
BM	=	3.032	kNm
Mu = 1.5 X BM	=	4.548	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (Mu + Defl.)	=	175.000	sqmm/m
Reinforcement Provided	=	T8 @ 285 C/C	
	=	176.000	sqmm/m
Short Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.040	
BM	=	4.042	kNm
Mu = 1.5 X BM	=	6.064	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	62.500	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m
Long Span Positive Moment At Midspan -			
Moment Coefficient	=	0.024	
BM	=	2.443	kNm
Mu = 1.5 X BM	=	3.665	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Area Of Reinforcement Required (BM + Defl.)	=	172.500	sqmm/m
Reinforcement Provided	=	T8 @ 290 C/C	
	=	173.000	sqmm/m
Long Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.032	
BM	=	3.258	kNm
Mu = 1.5 X BM	=	4.887	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of	=	63.750	sqmm/m

Reinforcement Required			
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.374	
V = SF1 x TL x Lx	=	9.723	kN
Vu = 1.5 X V	=	14.584	kN
Shear Stress, Tv	=	0.146	N/sqmm
Nominal Shear Stress, Tc	=	0.291	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.378	
	>	0.146	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.330	
V = SF2 x TL x Lx	=	8.570	kN
Vu = 1.5 X V	=	12.855	kN
Shear Stress, Tv	=	0.143	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.143	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.593	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.59 x 1.00 x 1.00	
	=	41.427	
Lx/d	=	39.200	
	<	41.427	Slab Is Safe In Deflection

Slab No. :	**S9**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.500	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Dead Load,DL	=	4.625	kN/sqm
Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	1-Way	
End Conditions	=	End Span	
Load Combination	=	1.5 DL + 1.5 LL	

Design Moments:

Short Span Positive Moment at Mid Span -

Moment Coefficient(DL), M1	=	0.083	
Moment Coefficient(LL), M2	=	0.100	
BM = M1 x DL x Lx x Lx + M2 x LL x Lx x Lx	=	1.099	kNm
Mu = 1.5 X BM	=	1.648	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Reinforcement -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Short Span Negative Moment at Continuous Support -

Moment Coefficient(DL), M3	=	0.100	
Moment Coefficient(LL), M4	=	0.111	
BM = M3 x DL x Lx x Lx + M4 x LL x Lx x Lx	=	1.285	kNm
Mu = 1.5 X BM	=	1.928	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

All Other Sections provide Minimum Steel @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Shear Coefficient(DL), SF1	=	0.400	
Shear Coefficient(LL), SF2	=	0.450	

V = SF1 x DL x Lx + SF2 x LL x Lx	=	5.446	kN
Vu = 1.5 X V	=	8.169	kN
Shear Stress, Tv	=	0.082	N/sqmm
Nominal Shear Stress, Tc	=	0.291	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.378	
	>	0.082	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26.000	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.516	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.52 x 1.00 x 1.00	
	=	39.414	
Lx/d	=	13.700	
	<	39.414	Slab Is Safe In Deflection

Slab No. :	**S10**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	3.920	m
Short Span, Lx	=	2.130	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	Two Adjacent Edges Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.067	
BM	=	2.013	kNm
Mu = 1.5 X BM	=	3.020	kNm

Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.089	
BM	=	2.684	kNm
Mu = 1.5 X BM	=	4.027	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.034	
BM	=	1.022	kNm
Mu = 1.5 X BM	=	1.533	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.045	
BM	=	1.363	kNm
Mu = 1.5 X BM	=	2.044	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.581	
V = SF1 x TL x Lx	=	8.196	kN
Vu = 1.5 X V	=	12.295	kN
Shear Stress, Tv	=	0.123	N/sqmm
Nominal Shear Stress, Tc	=	0.291	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.378	
	>	0.123	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.400	
V = SF2 x TL x Lx	=	5.645	kN
Vu = 1.5 X V	=	8.467	kN
Shear Stress, Tv	=	0.094	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.094	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	32	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.516	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	32.00 x 1.00 x 1.52 x 1.00 x 1.00	
	=	48.509	
Lx/d	=	21.300	
	<	48.509	Slab Is Safe In Deflection

Slab No. :	**S11**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	3.920	m
Short Span, Lx	=	2.130	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	One Long Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.064	
BM	=	1.916	kNm
Mu = 1.5 X BM	=	2.873	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.085	
BM	=	2.554	kNm
Mu = 1.5 X BM	=	3.831	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.028	
BM	=	0.827	kNm
Mu = 1.5 X BM	=	1.240	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.037	
BM	=	1.102	kNm
Mu = 1.5 X BM	=	1.653	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.564	
V = SF1 x TL x Lx	=	7.965	kN
Vu = 1.5 X V	=	11.948	kN
Shear Stress, Tv	=	0.119	N/sqmm
Nominal Shear Stress, Tc	=	0.291	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.378	
	>	0.119	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	5.080	kN
Vu = 1.5 X V	=	7.620	kN
Shear Stress, Tv	=	0.085	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.085	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	32	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.516	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	32.00 x 1.00 x 1.52 x 1.00 x 1.00	
	=	48.509	
Lx/d	=	21.300	
	<	48.509	Slab Is Safe In Deflection

Slab No. :	**S12**		
Level	=	5.029 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	2.130	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	125.000	mm
Effective Depth Along LX, Deffx	=	100.000	mm
Effective Depth Along LY, Deffy	=	90.000	mm
Self Weight	=	3.125	kN/sqm
Total Load, TL	=	6.625	kN/sqm
Span	=	2-Way	
Panel Type	=	Two Adjacent Edges Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.060	
BM	=	0.748	kNm
Mu = 1.5 X BM	=	1.121	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.080	
BM	=	0.997	kNm
Mu = 1.5 X BM	=	1.495	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.034	
BM	=	0.423	kNm
Mu = 1.5 X BM	=	0.634	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.045	
BM	=	0.564	kNm
Mu = 1.5 X BM	=	0.846	kNm
Area Of Reinforcement Required (Mu)	=	150.000	sqmm/m
Additional Area Of Reinforcement Required	=	75.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	120.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.547	
V = SF1 x TL x Lx	=	4.961	kN
Vu = 1.5 X V	=	7.441	kN
Shear Stress, Tv	=	0.074	N/sqmm
Nominal Shear Stress, Tc	=	0.291	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.378	
	>	0.074	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.400	
V = SF2 x TL x Lx	=	3.631	kN
Vu = 1.5 X V	=	5.446	kN
Shear Stress, Tv	=	0.061	N/sqmm
Nominal Shear Stress, Tc	=	0.305	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.397	
	>	0.061	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	32	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.516	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	32.00 x 1.00 x 1.52 x 1.00 x 1.00	
	=	48.509	
Lx/d	=	13.700	
	<	48.509	Slab Is Safe In Deflection